Stroma World — Minimal Gene Set v1.0

> **Constraint:** Smallest set that captures PDAC "personality" at the tumor-stroma interface. **Total: 21 genes.** Each earns its seat by directly mapping to a PhysiCell phenotype parameter or a Boolean network node. If a gene doesn't change a cell's behavior in simulation, it's cut.

Selection Logic

PDAC has four canonical driver events that appear in >50% of cases: **KRAS** gain-of-function, **CDKN2A** loss, **TP53** loss, **SMAD4** loss. These form the genetic chassis. Everything else is selected because it's either (a) downstream of these four and controls a phenotype axis we simulate, or (b) mediates the tumor↔stroma crosstalk that *is* the stroma barrier.

Genes are tagged by which cell type expresses them in our simulation:

- **[T]** = tumor cells (PDAC)
- **[S]** = stromal cells (PSCs / CAFs)
- **[T/S]** = both

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1. Growth Drive (proliferation rate)

#	Gene	Cell	Role in Stroma World
1	**KRAS**	[T]	Constitutively active in ~95% PDAC. Master ON switch for proliferation, survival, and secretion. The root node of the tumor Boolean network.
2	**MYC**	[T]	Transcriptional amplifier downstream of KRAS. Proxy for "how hard is the tumor pushing on growth." Maps directly to proliferation rate scaling.

#	Gene	Cell	Role in Stroma World
3	**EGFR**	[T]	Receptor input node. Allows the EA to test interventions that block upstream receptor signaling. Needed to give the optimizer a druggable handle on growth.

> **Why only 3?** Growth drive in PDAC is dominated by constitutive KRAS. Adding more RTKs or cyclins doesn't change simulation behavior — KRAS is always ON, MYC amplifies, EGFR is the druggable entry point.

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2. Death Resistance (apoptosis rate)

#	Gene	Cell	Role in Stroma World
4	**TP53**	[T]	Lost/mutant in ~75% PDAC. When OFF, apoptosis rate drops and DNA-damage-induced death is disabled. Boolean gate: TP53=OFF → apoptosis threshold ↑.
5	**BCL-XL** (BCL2L1)	[T]	Anti-apoptotic effector. Downstream of KRAS/NFκB. The node that actually suppresses the death phenotype. Druggable (navitoclax family).
6	**BAX**	[T]	Pro-apoptotic effector. Antagonist to BCL-XL. Needed to create the apoptosis toggle: if BCL-XL > BAX → cell lives. Simple ratio maps to apoptosis probability.

> **Why BAX and not full BCL-2 family?** One pro- and one anti-apoptotic member is the minimum to create a tunable death switch. More granularity adds parameters without changing the qualitative behavior.

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3. Cell-Cycle Braking (growth arrest / senescence)

#	Gene	Cell	Role in Stroma World
7	**CDKN2A** (p16)	[T]	Lost in ~90% PDAC. The primary cell-cycle brake. When OFF, the tumor ignores contact inhibition and crowding signals. Boolean gate for density-dependent arrest.
8	**SMAD4**	[T]	Lost in ~55% PDAC. TGF-β signal transducer. When intact, TGF-β = growth arrest. When lost, TGF-β still activates stroma but *cannot* brake the tumor. This asymmetry is critical to the stroma barrier problem.
9	**RB1**	[T]	Downstream integrator of cell-cycle checkpoints. Needed as the convergence node for CDKN2A and other brake signals in the Boolean network.

> **§2.2 link:** SMAD4 loss is what makes TGF-β a purely stromagenic signal in late PDAC. This gene *must* be in the network or the stroma barrier doesn't emerge correctly.

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4. Invasion & Motility (speed, adhesion, ECM remodeling)

#	Gene	Cell	Role in Stroma World
10	**ZEB1**	[T]	EMT master regulator. When ON → motility ↑, adhesion ↓. The single switch for mesenchymal phenotype. Downstream of TGF-β and hypoxia.
11	**CDH1** (E-cadherin)	[T]	Adhesion molecule. ZEB1 represses CDH1. CDH1=OFF is the operational definition of EMT in the sim. Maps to cell-cell adhesion strength parameter.
12	**MMP2**	[T/S]	Matrix metalloproteinase. Degrades ECM. Both tumor and CAFs secrete it. Maps to local ECM density reduction rate. This is the stroma-degrading effector the EA needs.

#	Gene	Cell	Role in Stroma World
13	**ACTA2** (α-SMA)	[S]	Marker/effector of CAF activation. When ON, the stellate cell is a myofibroblast: contractile, ECM-producing, motile. The "activated CAF" phenotype switch.

> **Why ZEB1 and not SNAIL/TWIST?** One EMT-TF is sufficient for the Boolean toggle. ZEB1 is the most stroma-responsive (TGF-β driven) and maps cleanly to a single motility switch.

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5. Secretion & Communication (tumor↔stroma crosstalk)

This is the axis that *makes* the stroma barrier. These genes define the paracrine loops.

#	Gene	Cell	Role in Stroma World
14	**TGFB1**	[T/S]	The primary stromagenic cytokine. Tumor secretes → activates PSCs → PSCs produce ECM and more TGF-β. Maps to TGF-β secretion rate. The positive feedback loop that builds the wall.
15	**SHH**	[T]	Sonic Hedgehog. Second paracrine arm. Tumor secretes → stroma responds via Hedgehog pathway → fibroblast proliferation + ECM. Druggable (vismodegib).
16	**GLI1**	[S]	SHH signal transducer in stromal cells. When ON → proliferation ↑, ECM production ↑. Needed so the EA can test SHH pathway inhibition at different nodes.
17	**HAS2**	[S]	Hyaluronan synthase. Produces hyaluronic acid, a major component of PDAC stroma that ↑ interstitial pressure and ↓ drug diffusion. Maps directly to ECM density field production rate. Druggable (PEGPH20/hyaluronidase).
18	**COL1A1**	[S]	Collagen I production. The structural backbone of the stromal barrier. CAFs with ACTA2=ON secrete COL1A1 at high rate → ECM density ↑ → drug diffusion ↓.

> **Why both HAS2 and COL1A1?** They represent two mechanistically distinct components of the barrier (gel-like HA vs. fibrillar collagen) with different druggable handles. PEGPH20 targets HA; collagenase/LOX inhibitors target collagen crosslinking. The EA needs both knobs.

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6. Stress Adaptation (survival under therapy / hypoxia)

#	Gene	Cell	Role in Stroma World
19	**HIF1A**	[T/S]	Hypoxia sensor. Dense stroma → poor perfusion → hypoxia → HIF1A ON → VEGF, metabolic shift, motility. Links stroma density to tumor adaptation. Boolean input: O_2 < threshold → ON.
20	**NRF2** (NFE2L2)	[T]	Oxidative/xenobiotic stress response. When ON → drug efflux ↑, ROS tolerance ↑. Represents the "stress-hardened" tumor phenotype that survives therapy. Maps to drug sensitivity parameter.
21	**ABCB1** (MDR1)	[T]	Drug efflux pump. Downstream of NRF2 and hypoxia. The direct mechanistic reason drugs fail even when they penetrate stroma. Maps to intracellular drug concentration decay rate.

> **Why include stress if this is a stroma-focused cycle?** Because the fitness function penalizes "drug reaches tumor but tumor survives anyway." Without stress adaptation, the EA will find solutions that open the stroma but don't actually kill anything — and you won't discover that failure until Cycle 2.

Summary Map

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GROWTH        DEATH        BRAKING
KRAS ──→ MYC   TP53 ──┐     CDKN2A ──→ RB1
EGFR ─┘        BCL-XL ←→ BAX  SMAD4 ─┘
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                    ↑
CROSSTALK           |
TGFB1 ←———————— [tumor] ——→ SHH    |
 ↓              ↓    |
SMAD4(lost)         GLI1    |
 ↓              ↓    |
[stroma]          [stroma]  |
ACTA2 → COL1A1       HAS2    |
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INVASION            STRESS    |
ZEB1 ——→ CDH1(off)    HIF1A ——→ NRF2 ——→ ABCB1
MMP2 (ECM degradation)
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Mapping to PhysiCell Phenotype Parameters

Behavioral Axis	Genes	PhysiCell Parameter(s)
Growth drive	KRAS, MYC, EGFR	cycle.transition_rate
Death resistance	TP53, BCL-XL, BAX	death.apoptosis_rate
Cell-cycle braking	CDKN2A, SMAD4, RB1	cycle.arrest flag, contact inhibition threshold
Invasion/motility	ZEB1, CDH1, MMP2, ACTA2	motility.speed, mechanics.cell_cell_adhesion, ECM degradation rate
Secretion	TGFB1, SHH, GLI1, HAS2, COL1A1	secretion.rate per substrate, ECM production rate
Stress adaptation	HIF1A, NRF2, ABCB1	Drug sensitivity modifier, O_2 response threshold

What's Deliberately Excluded

Excluded Gene/Pathway	Why
VEGF / angiogenesis	No vasculature dynamics in Cycle 1 (§2.3 of spec)
PD-L1 / immune checkpoints	No immune cells in Cycle 1
BRCA1/2, ATM (DNA repair)	Relevant for therapy response but not stroma barrier formation
Wnt / Notch	Important in differentiation but don't control stroma-barrier phenotypes directly
Full metabolic network	Cycle 3 scope; Warburg effect approximated via HIF1A
Additional MMPs (MMP9, MMP14)	MMP2 is sufficient as a single ECM-degrading proxy
PDGF / FGF	Additional stromal mitogens; GLI1 + TGFB1 already cover CAF activation

21 genes. Each one maps to a simulation parameter. Each one is either a PDAC driver, a stroma-barrier builder, or a druggable handle the EA needs. Nothing decorative.